# GPU Waste Detection for Kubernetes: Market Research, Competition, Unit Economics & International Expansion

Scope: the product **as it is actually built today** (September 2026), not the broader vision. That product is a read-only, self-hosted Helm install that reads NVIDIA DCGM telemetry plus Kubernetes pod identity, classifies GPU waste into four categories (idle, over-provisioned, fragmented, recurring pattern), prices each finding in INR against a right-sizing ceiling, and shows it on a dashboard. It recommends; it does not act. It has been validated on GKE (L4), on a vast.ai/k3s GPU node, against the real Microsoft Philly training trace (23.0% of GPU-workload-hours flagged as waste) and the real Azure LLM inference trace (19,366/19,366 requests served, over-provisioning finding fired).

How to read the numbers: every figure is tagged **[S]** (sourced, see Sources at the end) or **[A]** (our assumption, listed in the Assumptions ledger). Market-research firm estimates vary widely and are treated as directional. Currency: INR; USD conversions at ₹88/USD [A].

## Executive summary

| Question | Answer |
|---|---|
| TAM | **Global:** ~$0.45B (2026) spent on cost visibility/optimization for GPU workloads on Kubernetes [S+A]. **India:** ~₹78-190 Cr/yr (~$9-22M) at this product's own price points [A], growing fast with India's GPU build-out. |
| SAM | **India, Kubernetes-based GPU fleets reachable with read-only, self-hosted, INR-priced software:** ~885 organizations, ~₹38 Cr/yr (~$4.4M) [A]. |
| SOM (3 yrs) | **India only: 27-45 paying customers, ~₹0.8-1.35 Cr ARR (~$92-153k)** [A]. Viable as a bootstrapped business; not venture-scale on this product alone. |
| International | Adding SE Asia, Middle East, US and EU raises TAM to **~₹860 Cr (~$98M)** and SAM to **~₹270 Cr (~$31M)** [A]. With the same 45 customers (30 India + 15 abroad), year-3 ARR rises from ₹1.35 Cr to **~₹2.6 Cr** and net profit from ~₹62 L to **~₹1.13 Cr** [A]. Overseas customers pay ~3-5x more for similar support effort. |
| Main competitor | CAST AI ($1B valuation, Jan 2026) and ScaleOps ($800M+, Mar 2026) automate; Datadog launched GPU Monitoring (Apr 2026) and overlaps most directly on detection. You win only where read-only + self-hosted + INR + non-hyperscaler clusters matter. |
| Unit economics | Gross margin 78-89% (no hosting cost; you're self-hosted). LTV:CAC 3.4x (Starter) to 18x (Enterprise); CAC payback 4-6 months, founder-led [A]. As a solo founder, break-even is ~2 customers with no salary, ~5 with a ₹1 L/month draw [A]; your time (~25 customers supportable alone) is the real limit, not money. |
| Biggest risk | **One-time-audit churn:** a customer finds the waste, fixes it, cancels. If churn doubles, Starter LTV:CAC drops to 1.7x (unprofitable). Also: GPU-count pricing underprices H100 fleets and overprices cheap-GPU fleets. |

## 1. Market context

| Signal | Figure | Tag |
|---|---|---|
| India end-user public cloud spending (Gartner) | $17.5B in 2026, up 28.1% from $13.7B in 2025 | [S] |
| India data-center GPU market (MarkNtel) | ~$0.17B (2025) to $5.29B (2032), 63% CAGR | [S] |
| IndiaAI Mission compute | 38,000+ GPUs onboarded at ~₹65/GPU-hr; target 100,000 by Dec 2026 | [S] |

| Signal | Figure | Tag |
|---|---|---|
| Indian GenAI startups (Nasscom) | 890+ in H1 CY2025 (740+ application layer); $990M cumulative funding | [S] |
| India GPU installed base (third-party survey) | 80,000+ GPUs across public/private deployments | [S, 3rd-party blog] |
| Global GPU-as-a-Service (MarketsandMarkets) | $8.21B (2025) to $26.62B (2030), 26.5% CAGR | [S] |
| Global Kubernetes cost management market | ~$1.75B (2025) to ~$2.23B (2026), ~27% CAGR; other estimates $1.38B (2025) | [S, analyst range] |
| Kubernetes for AI inference (CNCF 2025 survey) | 66% of orgs hosting GenAI use Kubernetes for some or all inference | [S] |
| GPU share of compute cost (Datadog) | GPU instances = 14% of compute costs across its customer base | [S] |

What this means: the value pool (GPU spend) is growing 25-60% a year, and Kubernetes is now the default place GPU inference runs. But the software pool this product can capture is a thin slice of GPU spend: cost-visibility tools typically capture 1% or less of the spend they watch.

# 2. TAM, SAM, SOM

## 2.1 Global TAM (top-down, for context)

Global Kubernetes cost-management spend is ~$2.23B in 2026 [S]. We assume 15-25% of it is GPU-driven [A], anchored on Datadog's finding that GPUs are already 14% of compute cost and rising [S], and on GPU nodes costing several times more per hour than CPU nodes. That gives **~$0.33-0.56B, midpoint ~$0.45B (~₹3,900 Cr)** of global spend on cost visibility and optimization for GPU workloads on Kubernetes. This is the category CAST AI, ScaleOps, Kubecost and Datadog are fighting over. It is **not** reachable by a two-person, INR-priced, founder-led product in the next three years; it matters as the ceiling and as the reason incumbents will keep investing here.

## 2.2 India TAM (bottom-up, at this product's own prices)

| Segment (orgs running dedicated NVIDIA GPU fleets, 2026) | Count | Basis |
|---|---|---|
| AI / GenAI startups that train or self-host models | ~900 | [A] from Nasscom's 890+ GenAI startups [S]: most of the ~150 infra/model-layer firms, a minority of the 740 app-layer firms (most call APIs), plus non-GenAI AI firms (vision, speech, health AI) |
| Enterprises with in-house GPU clusters (BFSI, telecom, IT services, e-commerce, auto) | ~600 | [A] |
| Research, academic and government labs (IITs, IISc, IndiaAI compute users) | ~300 | [A] |
| GPU clouds / neoclouds / MSPs (E2E, Yotta, Neysa, NxtGen, Netweb, smaller) | ~30 | [A] better treated as channel partners than end customers |
| **Total** | **~1,800** | |

| Tier (current pricing) | GPUs | Annual price | Assumed share of orgs [A] |
|---|---|---|---|
| Starter | 1-5 | ₹1.2 L | 45% |
| Growth | 6-25 | ₹3.6 L | 35% |
| Scale | 26-100 | ₹9.0 L | 15% |
| Enterprise (assumed ₹2 L/mo) | 100+ | ₹24 L | 5% |

| Tier (current pricing) | GPUs | Annual price | Assumed share of orgs [A] |
|---|---|---|---|
| **Blended ACV at list** | | **₹4.35 L** | |

**Method A (by organization):** 1,800 orgs x ₹4.35 L = **~₹78 Cr/yr (~$8.9M)**. **Method B (by GPU):** 80,000 GPUs [S, third-party] x ~₹2,000 per GPU-month (the Growth tier's effective per-GPU price at ~15 GPUs) x 12 = **~₹192 Cr/yr (~$21.8M)**. India TAM is therefore **~₹78-190 Cr/yr**. Method B is higher because it counts GPU-cloud and IndiaAI capacity that sits in a few very large fleets. With IndiaAI alone targeting 100,000 GPUs by Dec 2026 [S], this number should roughly double by 2027-28.

## 2.3 SAM (India, serviceable by the product as built)

The product requires Kubernetes, NVIDIA GPUs with DCGM, and a buyer willing to install a Helm chart. Filters applied [A]:

- Exclude GPU clouds as direct customers (channel instead): 1,770 end-user orgs.
- Keep only fleets run on Kubernetes: 50% [A]. This is deliberately below CNCF's 66% [S] because many Indian startups run GPUs on bare VMs, Slurm, or notebooks rather than Kubernetes.
- Result: **~885 orgs x ₹4.35 L = ~₹38.5 Cr/yr (~$4.4M)**.

Not filtered out, but harder to win inside the SAM: teams on EKS/GKE/AKS that already have CAST AI, Datadog, or hyperscaler cost tools, and teams whose GPU spend is subsidized (IndiaAI at ~₹65/GPU-hr shrinks the rupee value of the same waste roughly 4-5x vs. commercial A100 rates).

## 2.4 SOM (3-year obtainable)

| Scenario | Share of SAM | Paying customers | Realized ACV [A] | ARR |
|---|---|---|---|---|
| Conservative | 3% | 27 | ₹3.0 L | ₹0.81 Cr (~$92k) |
| Base | 4% | 35 | ₹3.0 L | ₹1.05 Cr (~$119k) |
| Stretch | 5% | 45 | ₹3.0 L | ₹1.35 Cr (~$153k) |

Realized ACV is below the ₹4.35 L list blend because early customers skew small and pilots get discounted [A]. 3-5% of SAM in three years is normal for a founder-led product with no sales team.

**Honest read:** the product as built supports a profitable small business (break-even for a solo founder is 2-5 customers, section 4), not a venture-scale one. Getting to venture scale needs one or more of: (1) selling outside India (SE Asia, Middle East, US, EU: a ~$26M international SAM, see section 5); (2) white-labelling through Indian GPU clouds to reach their customers; (3) moving from detection to action (the queue-depth/TTFT and bin-packing items from the MVP roadmap), which is what justifies CAST AI's and ScaleOps's valuations.

## 3. Competitive landscape

Four kinds of competitor: **automation platforms** that change your cluster (CAST AI, ScaleOps, PerfectScale), **cost-allocation tools** (Kubecost/OpenCost), **observability platforms** adding GPU views (Datadog), and **schedulers** (NVIDIA Run:ai / KAI). The default competitor is still **DIY**: dcgm-exporter + Prometheus + Grafana.

| Competitor | What it is | GPU capability | Pricing (public / reported) | Scale | Overlap |
|---|---|---|---|---|---|
| **CAST AI** | Kubernetes automation platform (autoscaling, bin-packing, Spot, rightsizing) | GPU node autoscaling, Spot GPUs, time-slicing/MIG sharing, GPU cost monitoring, OMNI Compute / GPU marketplace | Free monitoring; paid tiers reported at $200 / $1,000 / $5,000 per month; savings-share ~15-20% reported (third-party) | $1B valuation (Jan 2026); ~$181M raised | High on visibility; they act, you recommend |
| **ScaleOps** | Autonomous real-time Kubernetes resource management | Adjusts compute and GPU resources automatically | Not public (enterprise) | $130M Series C at $800M+ (Mar 2026); $210M total; 350%+ YoY growth | Medium-high |
| **Datadog GPU Monitoring** | Observability platform add-on (GA Apr 22, 2026) | Fleet health, idle/misconfigured GPUs, cost + chargeback by team, cloud / neocloud / on-prem | Usage-based on top of Datadog agent pricing | Public company; huge installed base | **Highest on detection** |
| **IBM Kubecost / OpenCost** | Kubernetes cost allocation (acquired by IBM Sep 2024; OpenCost is the CNCF open-source core) | Per-GPU hourly cost allocation; idle-capacity cost; not GPU-utilization waste classification | Free/OSS tier; Business plan reported ~$449/mo per 100 nodes | IBM FinOps suite (Apptio, Cloudability, Turbonomic) | Medium |
| **PerfectScale (DoiT)** | Kubernetes optimization and governance (acquired by DoiT Feb 2025) | Rightsizing across CPU, GPU, storage, networking | Bundled with DoiT services | Part of DoiT | Medium |
| **NVIDIA Run:ai / KAI Scheduler** | GPU scheduler/orchestrator; KAI is the Apache 2.0 open-source core (CNCF Sandbox) | Fair-share quotas, gang scheduling, GPU sharing; utilization dashboards in Run:ai | KAI free; Run:ai enterprise | NVIDIA | Low (different layer) |
| **DIY: DCGM + Prometheus + Grafana** | Free open-source metrics stack | Raw utilization per GPU; no waste taxonomy, no workload rollups, no ₹ recommendations out of the box | Free (engineer time) | Everywhere | High on raw data |
| **Indian GPU clouds' own consoles** | E2E, Yotta, etc. instance dashboards | Basic per-instance utilization for their own GPUs | Included | Growing fast | Low; potential channel partners |

## 3.1 Feature comparison vs. CAST AI (closest well-known name)

| Capability | Your product | CAST AI |
|---|---|---|
| GPU waste detection (idle / over-provisioned / fragmented / recurring) | Yes, four categories, confidence-scored | Partial (utilization + cost monitoring; not this taxonomy) |
| ₹ savings estimate per finding | Yes, native INR, SLA-aware right-sizing ceiling | USD cost; savings reported after automation |
| Automatic action (scale, rebin, Spot, sharing) | **No** (recommends only) | **Yes** (core product) |
| Cluster permissions | **Read-only** (get/list/watch only) | Needs write access for automation |
| Where data lives | **Customer's own cluster** (self-hosted, SQLite) | CAST AI SaaS |
| Cluster types | Any Kubernetes with NVIDIA+DCGM; tested GKE and k3s/vast.ai | Strongest on EKS/GKE/AKS |
| Cloud-billing integration | No | Yes |
| Serving metrics (queue depth, TTFT) | No (roadmap) | Partial |
| Multi-cluster, SSO, SOC 2 | No | Yes |
| Pricing model | Flat INR subscription by GPU count | USD tiers + savings share |

## 3.2 Where you win and where you lose

| You win when... | You lose when... |
|---|---|
| - Security won't grant write access to a third-party agent (banks, government, regulated health)<br>- Data must stay in-cluster or in-country (DPDP Act, sovereign-AI projects)<br>- The cluster is on-prem, a neocloud, or k3s, not EKS/GKE/AKS<br>- The buyer wants a flat INR invoice, not a USD savings share<br>- The team is small (5-50 GPUs) and not a Datadog customer | - The buyer wants savings to happen automatically (CAST AI, ScaleOps)<br>- They already pay Datadog and can switch on GPU Monitoring<br>- They need billing reconciliation, multi-cluster, SSO, SOC 2<br>- Their GPUs are subsidized and the waste isn't worth ₹ to them<br>- An engineer says "we'll build it in Grafana" |

**Defensibility is low today.** The detection logic can be copied, and Datadog shipped a similar capability this year. Your advantages are distribution (India, INR, founder-led), the trust model (read-only, self-hosted), and speed. The durable moat has to come later from real pilot data (calibrated thresholds, per-workload baselines) and from being the tool Indian GPU clouds bundle.

## 4. Unit economics

The product is self-hosted in the customer's cluster, so you pay nothing for compute or storage per customer. Cost of goods is mostly **your support time**. All inputs below are assumptions [A] until pilots produce real data; founder time is costed at ₹1,500/hour [A].

### 4.1 Per-customer, monthly

| Line item | Starter | Growth | Scale | Enterprise |
|---|---|---|---|---|
| Price / month | ₹9,999 | ₹29,999 | ₹74,999 | ₹2,00,000 [A] |
| Hosting / compute | ₹0 | ₹0 | ₹0 | ₹0 |
| Shared tooling (CI, registry, email, domain) | ₹500 | ₹500 | ₹500 | ₹500 |
| Support hours / month [A] | 1 h = ₹1,500 | 2 h = ₹3,000 | 5 h = ₹7,500 | 12 h = ₹18,000 |
| Payment processing (2%) [A] | ₹200 | ₹600 | ₹1,500 | ₹4,000 |
| **COGS** | ₹2,200 | ₹4,100 | ₹9,500 | ₹22,500 |
| **Gross profit / month** | **₹7,799 (78%)** | **₹25,899 (86%)** | **₹65,499 (87%)** | **₹1,77,500 (89%)** |

### 4.2 Acquisition, lifetime value, payback

| Metric | Starter | Growth | Scale | Enterprise |
|---|---|---|---|---|
| CAC: founder-led sales, incl. ~4 free scans per win [A] | ₹40,000 | ₹90,000 | ₹2,50,000 | ₹6,00,000 |
| Onboarding (install, 7-day scan, readout) [A] | 4 h = ₹6,000 | 8 h = ₹12,000 | 16 h = ₹24,000 | 40 h = ₹60,000 |
| **Total acquisition cost** | ₹46,000 | ₹1,02,000 | ₹2,74,000 | ₹6,60,000 |
| Monthly churn [A] | 5% | 3% | 2% | 1.5% |
| Expected lifetime | 20 months | 33 months | 50 months | 67 months |
| **LTV** (gross profit x lifetime) | ₹1.56 L | ₹8.63 L | ₹32.7 L | ₹1.18 Cr |
| **LTV : CAC** | **3.4x** | **8.5x** | **12.0x** | **17.9x** |
| CAC payback | 5.9 months | 3.9 months | 4.2 months | 3.7 months |
| LTV : CAC if churn doubles | 1.7x | 4.2x | 6.0x | 9.0x |

Rule of thumb: LTV:CAC above 3x and payback under 12 months is healthy. Every tier clears that on paper, but **Starter is marginal**: it breaks if churn is higher than assumed, and it breaks as soon as a paid salesperson replaces founder time. CAC is founder-time CAC; hiring an account executive at ~₹1.5-2 L/month [A] roughly doubles to triples CAC for the lower tiers.

### 4.3 Customer-side ROI: does the price make sense to the buyer?

Recoverable waste assumed at **10% of GPU spend** [A]. Basis: the Philly replay flagged 23.0% of GPU-workload-hours as waste [S, own test]; not all flagged waste is recoverable (SLA headroom, team inertia), so we assume 30-50% of it gets acted on, i.e. ~7-12%. GPU rates are the product's own illustrative INR defaults (L4 ₹70/h, A100 80GB ₹300/h, H100 SXM ₹520/h), and IndiaAI's ~₹65/h [S]; 730 hours/month.

| Customer fleet | GPU spend / month | Tier | Price as % of spend | 10% recovered | ROI (x price) |
|---|---|---|---|---|---|
| 4 x L4 | ₹2.04 L | Starter | 4.9% | ₹20,440 | **2.0x** (weak) |
| 16 x GPUs on IndiaAI subsidy | ₹7.59 L | Growth | 4.0% | ₹75,920 | **2.5x** (weak) |
| 4 x A100 80GB | ₹8.76 L | Starter | 1.1% | ₹87,600 | **8.8x** |
| 16 x A100 80GB | ₹35.0 L | Growth | 0.86% | ₹3.50 L | **11.7x** |
| 64 x H100 SXM | ₹2.43 Cr | Scale | 0.31% | ₹24.3 L | **32x** (underpriced) |

**Pricing problem:** pricing by GPU count ignores that an H100 costs ~7x an L4. Small fleets of cheap or subsidized GPUs get a weak 2-2.5x return (a 2x ROI is usually not enough for a buyer to bother), while large H100 fleets pay 0.3% of spend for a 32x return. **Recommendation:** price on monitored GPU spend, e.g. ~1% of monitored monthly GPU spend with a ₹9,999 floor and volume discounts [A]. That keeps ROI around 10x for everyone and grows revenue with the customer's fleet.

### 4.4 Company break-even (solo founder)

You are a solo founder doing engineering and sales, and the product has no hosting cost, so the company's fixed costs are small [A]:

| Monthly fixed cost | No salary | With ₹1 L/month personal draw |
|---|---|---|
| GPU test rentals (occasional) | ₹10-20k | ₹10-20k |
| CA / GST compliance | ₹5-10k | ₹5-10k |
| Tools, domain, email | ₹3-5k | ₹3-5k |
| Travel to customer meetings | ₹10-20k | ₹10-20k |
| Founder draw | ₹0 | ₹1,00,000 |
| **Total (rounded)** | **~₹50k** | **~₹1.5 L** |
| **Break-even customers** (avg ~₹31,300/month gross profit at the 45/35/15/5 tier mix) | **~2** | **~5** |

**Your time is the constraint, not money.** At ~2.5 support hours per customer per month and ~40% of a 160-hour month on support, one person can look after **~25 customers**. Winning one customer takes ~30 hours (about 4 free scans plus onboarding), so ~30 sales hours a month gives ~1 new customer a month, roughly the 3-year SOM. Two levers stretch this: make the free scan self-serve (cuts a scan from ~6 hours to ~1), and favour Growth tier and above. **First hire** (a support engineer, ~₹1.25 L/month): when support passes ~40% of your time or ~20-25 customers (~₹60-75 L ARR).

### 4.5 What would change these numbers most

- **Churn** (the biggest swing factor). If customers treat the product as a one-time audit, lifetime collapses. Counter-measures: continuous value (weekly ₹ reports, regression alerts when waste returns, per-team chargeback) and annual contracts.

- **Free-scan conversion.** CAC assumes ~1 in 4 free scans converts. At 1 in 10, lower-tier CAC more than doubles.
- **Pilot-proven savings.** The whole ROI table rests on the 10% recoverable assumption. The first pilot's before/after number replaces it.
- **Moving to action** (bin-packing recommendations, queue-depth-aware right-sizing) supports a higher price and lower churn, but brings you into direct competition with CAST AI and ScaleOps.

## 5. International expansion

Overseas buyers tend to run larger GPU fleets and pay in USD, so each customer is worth ~2-6x an Indian one for broadly similar support effort. That matters most for a solo founder, whose limit is customer count, not money. Four candidate regions:

| Region | Market signals [S] | Competition | Fit for this product |
|---|---|---|---|
| **SE Asia** (Singapore, Malaysia, Indonesia, Philippines) | 680+ AI startups, $2.3B AI funding in 12 months (e-Conomy SEA 2025); Malaysia leads SE Asia data-centre construction (1,039 MW); Aolani + NVIDIA plan 22,000 GPUs in Malaysia/Philippines | Lighter; global tools present but little local focus | **Good.** Close time zones, similar price sensitivity, new GPU clouds as channel partners |
| **Middle East** (UAE, Saudi Arabia) | MENA AI startups raised $858M in 2025 (60% UAE); 673 AI companies in Abu Dhabi; HUMAIN ordered 18,000 GB300 GPUs; G42's Stargate UAE first stage ~400,000 NVIDIA chips | Sovereign buyers work through large integrators; few SaaS tools allowed in | **Good for the trust model.** Read-only + self-hosted suits sovereign data rules, but needs a local partner; the mega-fleets are enterprise deals out of solo reach |
| **United States** | ~6,956 AI startups (largest share globally); neocloud revenue >$25B in 2025, projected ~$42B in 2026 | **Heaviest:** CAST AI, ScaleOps, Datadog, Kubecost all sell here | **Largest pool, hardest sale.** Target neocloud/on-prem GPU users, not EKS/GKE teams; SOC 2 required |
| **EU + UK** | ~1,192 GenAI companies in Europe; UK 800+ AI startups, France 468, Germany 394 | CAST AI (EU-founded), Datadog, Kubecost | **Moderate.** Data-residency rules favour self-hosted; GDPR paperwork; SOC 2 / ISO 27001 expected |

## 5.1 TAM, SAM, SOM by region (bottom-up, at this product's prices)

Pricing assumed at ~1% of monitored GPU spend in USD. Org counts and average contract values are our assumptions [A], anchored on the startup and funding counts above [S]. SAM keeps orgs on Kubernetes (CNCF: 66% of GenAI hosts [S]) and, in the US and EU, only the share that values a read-only, self-hosted tool or runs on neocloud/on-prem clusters.

| Region | Orgs with GPU fleets [A] | Avg contract / yr [A] | TAM | SAM filter [A] | SAM | 3-yr SOM [A] |
|---|---|---|---|---|---|---|
| India | ~1,800 | ₹3-4.35 L | ₹78 Cr ($8.9M) | 50% on K8s | ₹38 Cr ($4.4M) | 30 cust, ₹0.90 Cr |
| SE Asia | ~370 | $8k (₹7 L) | ₹26 Cr ($3.0M) | 55% on K8s | ₹14 Cr ($1.6M) | 6 cust, ₹0.42 Cr |
| Middle East | ~250 | $20k (₹17.6 L) | ₹44 Cr ($5.0M) | 55% on K8s | ₹24 Cr ($2.8M) | 3 cust, ₹0.53 Cr |
| United States | ~4,200 | $15k (₹13.2 L) | ₹554 Cr ($63M) | 66% K8s x 40% fit | ₹146 Cr ($16.6M) | 4 cust, ₹0.53 Cr |
| EU + UK | ~1,500 | $12k (₹10.6 L) | ₹158 Cr ($18M) | 60% K8s x 50% fit | ₹48 Cr ($5.4M) | 2 cust, ₹0.21 Cr |
| **Intl. only** | ~6,300 | | **₹783 Cr ($89M)** | | **₹233 Cr ($26.4M)** | **15 cust, ₹1.69 Cr** |
| **India + intl.** | ~8,100 | | **~₹860 Cr ($98M)** | | **~₹270 Cr ($31M)** | **45 cust, ~₹2.6 Cr** |

**Consistency check:** the bottom-up total (~$98M) is about 22% of the ~$446M global top-down TAM in section 2.1. The top-down figure also counts very large hyperscaler customers, automation platforms' revenue and spend this product can't serve, so a bottom-up number well below it is expected. **Why SOM is small abroad:** 15 overseas customers is 0.4-3% of each region's SAM, limited by one founder's selling time, not by demand.

## 6. Comparison: India only vs. India + international (year 3)

Same total of 45 customers in both cases, since one founder plus one or two support engineers can't handle many more. Figures in ₹ lakh per year; all costs are assumptions [A].

| | **India only (45 India)** | **India + international (30 India + 15 abroad)** |
|---|---|---|
| **Revenue (ARR)** | **135** | **259** |
| Payment processing (+ FX abroad) | 2.7 (2%) | 6.5 (2.5%) |
| CI, registry, tooling | 1.0 | 1.5 |
| Support engineers | 15 (one) | 30 (two; one covering US/EU hours) |
| **Gross profit** | **116 (86%)** | **221 (85%)** |
| SOC 2 / security certification | 4 | 20 (required by overseas buyers) |
| US company + US accountant | 0 | 3 |
| Travel | 2.4 | 12 (2-3 overseas trips) |
| Conferences (attend, not exhibit) | 0 | 5 |
| CA, legal, tools, GPU testing | 3.6 | 6 |
| **Profit before your salary** | **106 (79%)** | **175 (68%)** |
| Your salary (₹2 L/month) | 24 | 24 |
| Corporate tax (~25%) | 21 | 38 |
| **Net profit** | **62 (46%)** | **113 (44%)** |

| | **India only** | **India + international** |
|---|---|---|
| TAM / SAM | ₹78 Cr / ₹38 Cr | ~₹860 Cr / ~₹270 Cr |
| Avg revenue per customer | ₹3 L | ₹5.8 L |
| Competition | Light; INR pricing and local presence are real advantages | Heavy in the US; your edge narrows to read-only / self-hosted / neocloud |
| Fixed costs to enter | Minimal | SOC 2, US entity, travel, conferences, 2nd support engineer: ~₹50 L/yr extra |
| Sales cycle and trust | Shorter; in-person possible | Longer; security reviews; solo-founder risk questions |
| Can you stay solo? | Until ~25 customers | Needs a support hire earlier because of time zones |
| Risk if it goes badly | Low: costs stay small | Higher: ~₹50 L/yr of fixed costs before overseas revenue |

**Recommendation: sequence it, don't choose.** (1) Now to year 1: India only. Get pilot-proven savings and 2-3 written case studies; you need them anywhere. (2) Years 1-2: SE Asia and the Middle East, sold remotely and through GPU-cloud partners; close time zones and a trust model that suits sovereign buyers. (3) Years 2-3: US and EU, once SOC 2 and case studies exist, targeting neocloud and on-prem GPU users rather than EKS/GKE teams already on CAST AI. This keeps the ~₹50 L/yr of international fixed costs until revenue can pay for them.

## 7. Assumptions ledger

| # | Assumption | Value | Why / how to validate |
|---|---|---|---|
| A1 | USD/INR | ₹88 | Approximate 2025-26 rate |
| A2 | GPU share of Kubernetes cost-management spend | 15-25% | Datadog: GPUs 14% of compute cost and rising; GPU nodes cost more per hour |
| A3 | Indian orgs running dedicated NVIDIA GPU fleets | ~1,800 | Built from Nasscom startup counts + estimates; validate via GPU-cloud partner conversations |
| A4 | Share of those on Kubernetes | 50% | Below CNCF's 66%; ask every prospect |
| A5 | Tier mix of customers | 45/35/15/5 | Skewed small; replace with pipeline data |
| A6 | Enterprise price | ₹2 L/month | Custom tier; placeholder |
| A7 | Realized ACV | ₹3.0 L | Discounts + small-customer skew |
| A8 | Founder time value | ₹1,500/hour | Opportunity cost placeholder |
| A9 | Support / onboarding hours | See 4.1-4.2 | Measure on first pilots |
| A10 | CAC per tier | ₹40k to ₹6 L | Founder-led, ~4 free scans per win |
| A11 | Monthly churn | 1.5-5% | Biggest uncertainty; see 4.5 |
| A12 | Recoverable waste | 10% of GPU spend | From Philly 23% flagged x 30-50% acted on; prove in pilot |
| A13 | Monthly fixed costs (solo founder) | ₹50k-1.5 L | With or without a ₹1 L personal draw |
| A14 | 3-year SAM capture (India) | 3-5% | Typical for founder-led B2B infrastructure software |
| A15 | Overseas orgs with GPU fleets | SEA 370, ME 250, US 4,200, EU 1,500 | Anchored on startup counts in section 5; validate with regional partners |
| A16 | Overseas average contract | $8k-20k/yr | ~1% of monitored GPU spend; larger fleets abroad |
| A17 | Overseas SAM filters | 55-66% on K8s; 40-50% fit in US/EU | Share that values read-only, self-hosted or runs neocloud/on-prem |
| A18 | International entry costs | ~₹50 L/yr | SOC 2, US entity, travel, conferences, 2nd support hire |
| A19 | Solo support capacity | ~25 customers | 2.5 h/customer/month, 40% of time on support |

## 8. Sources


- Gartner, India public cloud forecast (Jun 2026): gartner.com/en/newsroom/press-releases/2026-06-01-gartner-forecasts-end-user-public-cloud-spending-in-india-to-surpass-17-billion-us-dollars-in-2026
- MarkNtel Advisors, India Data Center GPU Market 2025-2032 (via prior research).
- Nasscom, India Generative AI Startup Landscape 2025: nasscom.in/knowledge-center/publications/india-generative-ai-startup-landscape-2025-mapping-momentum
- PIB (Government of India), IndiaAI Mission compute update: pib.gov.in/PressReleasePage.aspx?PRID=2245069
- Introl, India GPU infrastructure survey (80,000+ GPUs): introl.com/blog/indias-gpu-infrastructure-landscape-a-comprehensive-survey
- MarketsandMarkets, GPU-as-a-Service market 2025-2030 (via prior research).
- Kubernetes cost management market (TBRC via Research and Markets / GII Research; GM Insights; Growth Market Reports): researchandmarkets.com/reports/6215668/kubernetes-cost-management-market-report
- CNCF 2025 Annual Cloud Native Survey (66% use K8s for GenAI inference): cncf.io/announcements/2026/01/20/
- Datadog GPU Monitoring GA (Apr 22, 2026): datadoghq.com/about/latest-news/press-releases/datadog-gpu-monitoring-launch/

- CAST AI $1B valuation, GPU marketplace (Jan 2026): siliconangle.com/2026/01/12/cast-ai-raises-funds-pacific-alliance-ventures-1b-valuation-launch-unified-gpu-marketplace/ ; multiples.vc/private-comps/cast-ai
- CAST AI pricing (third-party): usage.ai/blogs/finops/tools/cast-ai-pricing/
- ScaleOps $130M Series C (Mar 30, 2026): prnewswire.com/news-releases/scaleops-raises-130m-series-c-at-over-800m-valuation-...-302728822.html ; thenextweb.com/news/scaleops-130m-series-c
- IBM acquires Kubecost (Sep 17, 2024): techcrunch.com/2024/09/17/ibm-acquires-kubernetes-cost-optimization-startup-kubecost/
- Kubecost pricing (third-party): trustradius.com/products/ibm-kubecost/pricing
- DoiT acquires PerfectScale (Feb 2025): businesswire.com/news/home/20250203440309/en/
- NVIDIA open-sources KAI Scheduler: developer.nvidia.com/blog/nvidia-open-sources-runai-scheduler-to-foster-community-collaboration/
- Bain, Google, Temasek, e-Conomy SEA 2025 (680+ AI startups, $2.3B AI funding): bain.com/insights/e-conomy-sea-2025/
- SE Asia data-centre pipeline (Malaysia 1,039 MW under construction): technode.global/2026/05/12/asia-pacific-ai-driven-data-center-boom-reshapes-southeast-asia-landscape/ ; Aolani + NVIDIA 22,000 GPUs: technode.global/2026/09/22/aolani-nvidia-22000-gpu-ai-infrastructure-southeast-asia/
- MENA AI funding $858M in 2025 (MAGNiTT): enterpriseam.com/ksa/2026/02/15/mena-ai-funding-hits-usd-858-mn-in-2025-led-by-the-uae-and-saudi-arabia/ ; Abu Dhabi 673 AI companies: startupgenome.com/insights/abu-dhabis-rise-as-a-global-ai-and-startup-capital
- HUMAIN and G42 GPU deployments: restofworld.org/2026/gulf-ai-investment-nvidia-monopoly-blackwell/ ; presenc.ai/research/sovereign-ai-infrastructure-tracker-2026
- Neocloud revenue >$25B in 2025 (DCD) and ~$42B 2026 projection (Research and Markets): datacenterdynamics.com/en/news/neocloud-revenue-exceeds-25bn-in-2025/ ; researchandmarkets.com/reports/6225920/neocloud-global-market-report
- AI startup counts, US and Europe (StartUs Insights, StartupBlink): startus-insights.com/innovators-guide/how-many-ai-companies-are-there/ ; startupblink.com/blog/top-ai-startups/
- Own test results: testing/real-trace-replay/RESULTS.md (Philly, 23.0% flagged); testing/azure-inference-trace-replay/RESULTS.md; PROGRESS.md.

Market-research-firm figures differ substantially between publishers; treat them as ranges. Competitor pricing marked third-party was not confirmed on the vendor's own pricing page.